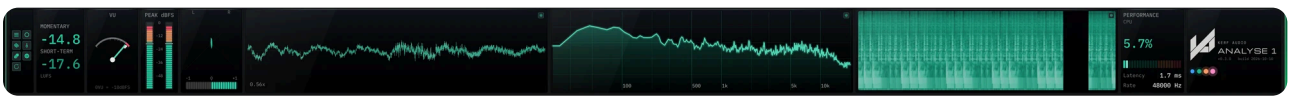
KERF AUDIO

USER MANUAL · VERSION 0.5.0 BETA

ANALYSE 1

A metering and analysis strip for REAPER. It shows what is leaving your master bus: loudness, level, stereo image, the waveform and the spectrum, all at once and all live.

MACOS 11+ | WINDOWS 10 AND 11 | LINUX X86_64 | REAPER 7.8 OR LATER | FREE



Free and in public beta. Analyse 1 has been through extensive code review and works on macOS, Windows and Linux. It hasn't yet been widely used in real-world projects, so please report anything that doesn't look right, especially a reading that disagrees with a meter you trust.

DANNY WINCOTT, TRADING AS KERF AUDIO · GITHUB.COM/KERF-AUDIO/ANALYSE-1

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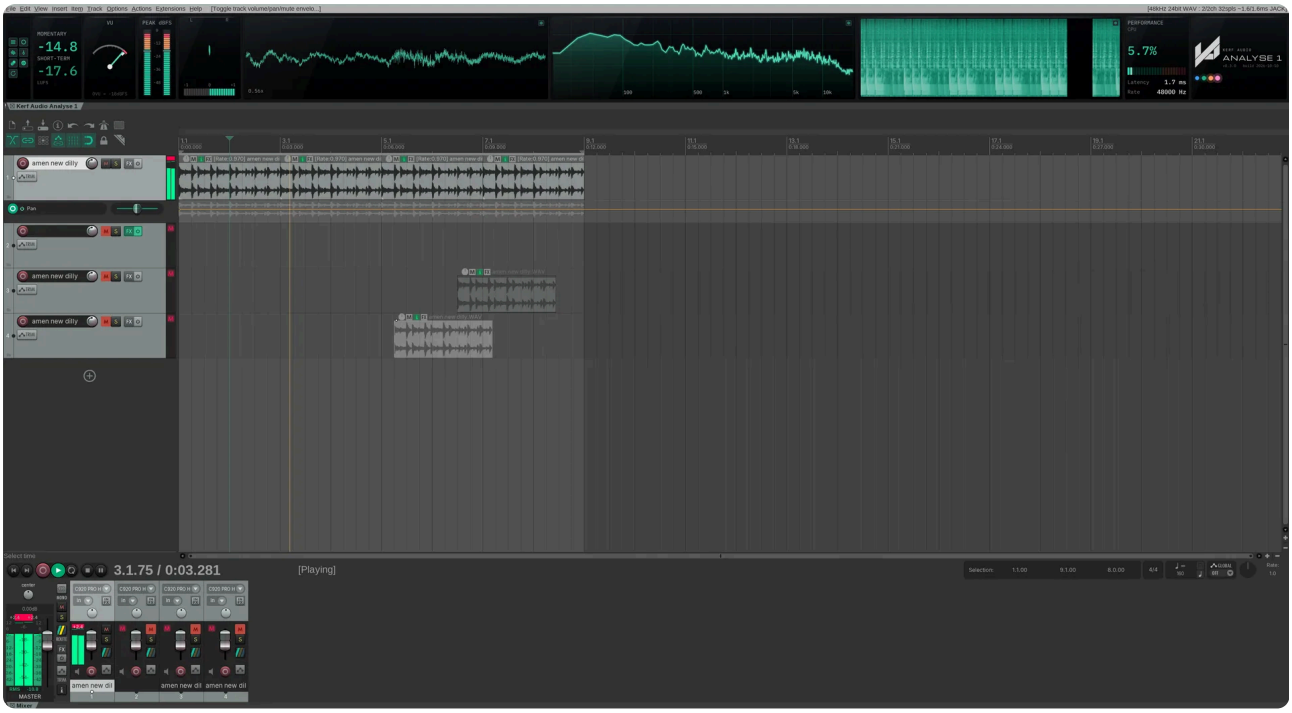
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WHAT IT IS

Analyse 1 sits in a docker along the top of REAPER's main window and shows you what is leaving your master bus.

There is no plugin to insert and nothing to arm. Analyse 1 listens to REAPER's output directly, so whatever you hear is what you see. Open the panel and it starts working.



ANALYSE 1 DOCKED AT THE TOP OF REAPER, MEASURING A DRUM LOOP

Modules

LUFS
Momentary and short-term loudness to ITU-R BS.1770-4 and EBU R128.

VU
A classic needle with 300 ms ballistics, 0 VU at −18 dBFS.

Peak
Fast left and right peak with RMS markers; dBFS, K-12, K-14 and K-20 scales.

Goniometer
A stereo vector scope with a phase-correlation meter.

Scope
An oscilloscope with zero-crossing trigger and zoom.

Spectrum
A live analyser from 20 Hz to 20 kHz on a fixed log scale, with hover readouts.

Spectrogram
A scrolling waterfall of the spectrum over time, with hover readouts.

Performance
CPU use, output latency and the device sample rate.

Brand
The Kerf Audio mark, with the exact version and build date you're running.

Features

Show, hide, reorder, resize
Choose which modules show, drag them into order and set each one's width.

Named layouts
Save, load and delete layouts by name.

Pop out
Put any module in its own resizable window, on a second screen if you like.

Freeze
Hold one module, or every module at the same instant, to read it properly.

Monitoring tools
Mono, LF solo and Dim on the listening path only, never in a render.

Master FX bypass
Compare with and without your master processing in one click.

Themes
Four accents, including one that follows your REAPER theme, in light or dark.

Zero setup, zero cost when off
No plugin to insert. Switch analysis off and it uses no processor time.

INSTALLING

Install with ReaPack, the free package manager for REAPER, from reapack.com. It works on every platform, keeps Analyse 1 up to date, and on macOS it is the only way to install.

SYSTEM	REQUIREMENT
macOS	macOS 11 Big Sur or later, Apple Silicon or Intel.
Windows	Windows 10 or 11, 64-bit.
Linux	64-bit x86 with glibc 2.35 or later. X11, or Wayland through XWayland.
REAPER	REAPER 7.8 or later, 64-bit.

1 **Add the Kerf Audio repository**

In REAPER, choose **Extensions › ReaPack › Import repositories...**, paste this address and click OK. Skip this if you added it for Kerf Modulators.

```
https://raw.githubusercontent.com/Kerf-Audio/reapack/main/index.xml
```

2 **Install Analyse 1**

Choose **Extensions › ReaPack › Browse packages**, search for **Analyse 1**, right-click it, choose **Install**, then **Apply**.

3 **Restart REAPER**

REAPER only looks for extensions when it starts.

Installing by hand (Windows and Linux only)

Download `reaper_analyser-x64.dll` (Windows) or `reaper_analyser-x86_64.so` (Linux) from github.com/Kerf-Audio/analyse-1/releases. Quit REAPER, copy the file into your REAPER `UserPlugins` folder (**Options › Show REAPER resource path in explorer/finder** shows where), then start REAPER. Make sure only one `reaper_analyser` file is in that folder.

Opening the panel

Open the **Extensions** menu and choose **Kerf Audio Analyse 1: Show/hide panel**. The menu item shows a tick while the panel is open. The same command is in REAPER's action list. On macOS it comes with the shortcut `Option+V`. On Windows and Linux it has no shortcut to begin with, so it can't clash with one you use; to add one, open **Actions › Show action list**, search for *Analyse 1* and press **Add** under Shortcuts. REAPER remembers whether the panel was open when you saved a project.

Where it sits

The strip is designed to live in a docker across the top of REAPER's main window, where it takes 150 pixels of height and leaves the arrange view alone. The first time it opens it moves itself to the top, and the docker height settles after REAPER's next restart. If you already have a docker at the top, Analyse 1 joins it. It is an

ordinary REAPER docker, so right-click its tab to dock it elsewhere or float it. Once you've moved it, Analyse 1 leaves your choice alone.

Updating and removing

ReaPack checks for updates on its own; apply them and restart REAPER. To try fixes early, right-click Analyse 1 in **Browse packages** and turn on pre-releases. To remove it, choose **Uninstall** there, or delete the file from `UserPlugins`, and restart. Nothing else is installed. Your layouts and settings stay in REAPER's settings file, and do no harm if you don't come back.

A TOUR OF THE STRIP



FROM THE LEFT: THE TOOLS COLUMN, LUFS, VU, PEAK, GONIOMETER, SCOPE, SPECTRUM, SPECTROGRAM, PERFORMANCE AND BRAND

Each instrument in the strip is a **module** with its own screen. You choose which modules show, their order and their width. On the far left is the tools column, with the menu and the monitoring controls.

Arranging modules

- **Show or hide** a module from the menu, under *Show / hide*.
- **Reorder** by dragging a module sideways to where you want it.
- **Resize** by dragging the gap between two modules. Only those two neighbours change.
- **Pop out** a module into its own window from the *Pop out* submenu. Close the window to send it back.
- **Save a layout** by name from *Layouts*, and load or delete saved layouts from the same place.

Every module has a smallest width it can draw at without cutting off its numbers. If the docker is too narrow for everything you've switched on, modules at the right-hand end step aside rather than everything being squashed. Widen the docker and they come back.

The tools column



The buttons have no labels, to save room. Hover over any of them to see its name.

BUTTON	WHAT IT DOES
Menu	Modules, layouts, monitoring and settings.
Mono	Folds what you hear to mono, to check the mix holds up on one speaker.
LF solo	Plays only the low end, in mono, below the crossover (60 to 200 Hz, 120 Hz to begin with; set it from *LF solo below*).
Dim	Turns what you hear down by 20 dB.
Freeze all	Holds every module still at the same instant.
FX bypass	Switches the master FX chain on and off, always showing REAPER's real state.
Reset holds	Clears anything held on screen, such as the CPU peak, and wipes the spectrogram.

> **Mono, LF solo and Dim only change what you hear.** They work on the monitoring path, after everything else, so they never end up in a render and never change your project. While one is on, an orange bar across the top of the strip says which, and how much it changes the sound. They switch themselves off when you hide the panel, so a hidden panel can never be quietly dimming your speakers.

The menu

- **Analysis enabled** switches Analyse 1 off completely. It stops listening and drawing, so it uses no processor time, while the panel stays where it is.
- **Theme**: four accents (Coral, Teal, Blue, and one that follows REAPER's own theme), in light or dark.
- The monitoring buttons, freeze, FX bypass and reset holds are repeated here, for when a short docker can't fit every button in the tools column.

04 · FREEZE

FREEZE

Freeze holds a display still so you can actually read it. Each module has its own small freeze button in its top right corner. **Freeze all** in the tools column freezes every module at the same moment, so the spectrum and the waveform you're comparing really are from the same instant. The audio carries on as normal; unfreeze and the module goes straight back to live.

THE MODULES

01 LUFS



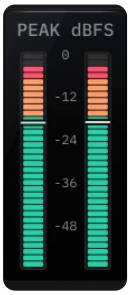
Momentary loudness (the last 400 ms) and short-term loudness (the last 3 seconds), measured to the ITU-R BS.1770-4 and EBU R128 standards. This is the loudness streaming services measure, so it's the one to watch when you're working to a target.

02 VU



A classic VU needle with proper 300 ms ballistics, calibrated so that 0 VU is a sine wave at −18 dBFS. It's slow on purpose: it shows the weight and body of the mix rather than every transient, which is why it's so useful next to a peak meter. The scale turns red above 0 VU.

03 Peak



Fast left and right peak meters, each with a marker showing the RMS level. The meter goes orange and then red as it nears full scale. Click the *PEAK* label to step through the scales: dBFS, then K-12, K-14 and K-20 for working to the K-System.

04 Goniometer



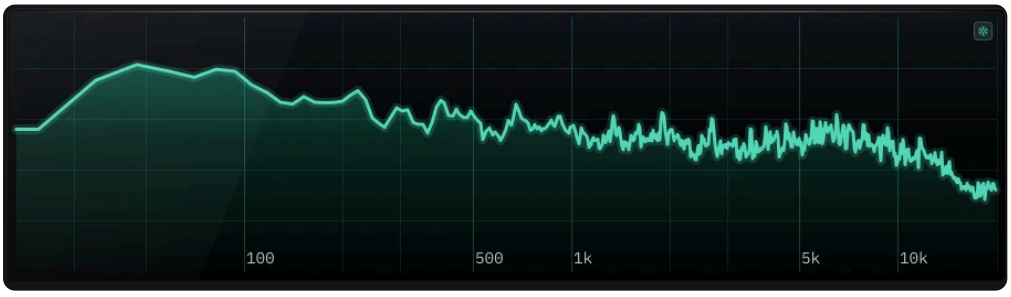
A stereo vector scope with a correlation meter underneath. A thin vertical line means mono, a wide cloud means a wide mix, and a trace leaning to one side means one channel is louder. The correlation bar reads from −1 to +1 and should mostly sit on the right. If it keeps lighting red on the left, part of your mix is out of phase and will lose level, or disappear, in mono.

05 Scope



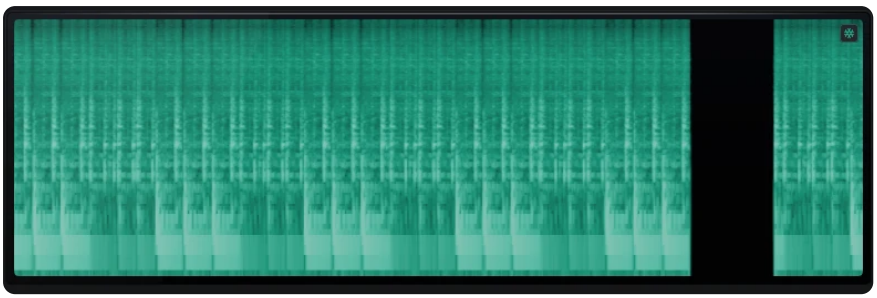
An oscilloscope that triggers on rising zero crossings, so a steady tone stands still instead of sliding across the screen. Scroll over it (or swipe on a trackpad) to zoom the time axis; the zoom is shown in the bottom left corner. The trace brightens as the signal gets louder.

06 Spectrum



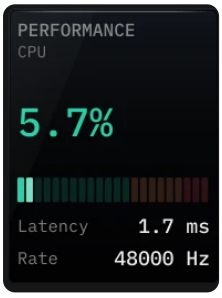
A live frequency analyser from 20 Hz to 20 kHz on a logarithmic scale. The scale never moves, so a frequency is always in the same place and you can compare sessions by eye. Hover to read the frequency, the nearest musical note and the level under the pointer. If your audio device runs at a low sample rate and can't reach 20 kHz, the part above its limit is shaded and the trace stops there.

07 Spectrogram



The spectrum over time, as a scrolling waterfall: newest on the right, low notes at the bottom, brighter for louder. Brilliant for spotting resonances, ringing tails and noise that flash past too quickly to catch on the spectrum. Hover to read frequency, note, level and how long ago that moment was. *Reset holds* clears it. It only records while the panel is on screen, so time spent hidden shows as a gap.

08 Performance



How hard REAPER is working: **CPU** use, with a meter that goes orange and red when things get busy, the output **Latency** of your audio device, and its sample **Rate**. Every measurement in Analyse 1 is based on that rate, so it's the first thing to check if a reading ever looks wrong.

09 Brand



The Kerf Audio mark, the product name, and the exact version and build date you're running. It quietly breathes with the music. If you report a problem, the version and build date shown here are the most helpful thing you can send. Hide it from the menu if you'd rather have the room.

06 · TROUBLESHOOTING

THINGS THAT MIGHT CROP UP

WHAT YOU SEE	WHY, AND WHAT TO DO
Not in the Extensions menu	REAPER only loads extensions when it starts, so quit it completely and start it again. Then check that exactly one `reaper_analyser` file is in `UserPlugins`, and that it's the right one: `.dylib` on macOS, `.dll` on Windows, `.so` on Linux.
Opened at the bottom, or the wrong height	The panel only moves itself to the top when it's sure it won't move another window's docker instead. Right-click the *Kerf Audio Analyse 1* tab, dock it at the top of the main window, then drag the docker edge until the strip looks right. REAPER remembers both.
The meters aren't moving	Check REAPER is playing out through your audio device; Analyse 1 reads REAPER's output. Check *Analysis enabled* is ticked in the menu, and that no module is frozen (a frozen module shows a lit freeze button).
Top of the spectrum shaded	Look at *Rate* in the Performance module. If it says 16000, your audio device is running at a low rate. The usual culprit is Bluetooth headphones switching to call mode when anything opens the microphone, which only carries sound up to 8 kHz and affects the loudness reading too. Use wired headphones or speakers for anything you intend to judge.
Quieter, mono or bass-heavy	Dim, Mono or LF solo is on; the orange bar says which. Click it again, or hide and show the panel, which switches all of them off.
Gaps in the spectrogram	Expected. It only records while the panel is visible.

07 · HOW IT WORKS

HOW IT WORKS

Analyse 1 hooks into REAPER's audio output at the very end of the signal chain, after the master bus and its effects, and copies what it hears into a buffer. That copy is all that happens on REAPER's audio thread. It never waits, never locks and never asks for memory, so Analyse 1 can't cause a dropout by being busy.

The measuring and drawing happen separately, about fifty times a second, on the same thread as REAPER's interface. A slow screen can never change a measurement, and when the panel is hidden or analysis is switched off, that work stops entirely.

HELP, BUGS AND THIS BETA

Analyse 1 is free, and a thank-you to the REAPER community, which has inspired me since 2007.

It has been through extensive code review and works on macOS, Windows and Linux. It's a beta because it hasn't yet been widely used in real-world projects, with all their different sample rates, channel counts and setups. That is where you come in.

Reporting a problem

Report bugs at github.com/Kerf-Audio/analyse-1/issues, and ask questions in Discussions on the same page. Support is through GitHub only. Every report is read, and fixes come as time allows. Please include:

- The version and build date from the Brand module.
- Your system and version, and your REAPER version.
- Your sample rate and channel count.
- What you were doing, what happened, and what you expected.
- If a reading looks wrong, the meter you compared it with.
- A screenshot, if the problem is something you can see.

> TESTING FIRST, FEATURES AFTER 1.0
>
> This first release phase is for testing and bug fixing. No new features will be added until version 1.0 is out and stable. Feature requests are welcome in Discussions or Issues on GitHub, and they will be looked at after 1.0.

Licence

Analyse 1 is free to use, including for commercial work, under the Kerf Audio Freeware Licence. It is not open source. You may not sell or redistribute it; link to the official download instead. It is provided as is, without warranty. The full licence and third-party notices are at github.com/Kerf-Audio/analyse-1. Analyse 1 uses Michroma and IBM Plex, both under the SIL Open Font Licence.

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KERF AUDIO

ANALYSE 1 · 0.5.0 BETA

Install with ReaPack, report bugs on GitHub, and tell us if a reading ever looks wrong.

```
https://raw.githubusercontent.com/Kerf-Audio/reapack/main/index.xml
```

github.com/Kerf-Audio/analyse-1